

Reliability-Aware Transfer Learning for Robust, Calibrated, and Explainable UAV-Based Aquatic Plant Classification

Leakage-Controlled UAV Aquatic Plant Dataset with ResNet50 and EfficientNet-B0

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1. Abstract

Unmanned Aerial Vehicle (UAV) imagery can support efficient wetland monitoring by recognizing aquatic plants. However, high clean-set accuracy does not guarantee reliable deployment. This study presents a reliability-aware transfer learning framework for the classification of 14 aquatic plants. This framework collectively considers data integrity, robustness, confidence calibration, selective prediction, explainability, and residual failure behavior. An audit of the original 1,468-image classification dataset revealed complete exact overlap between the supplied training and test partitions. After duplicate removal and near-duplicate-aware grouping, 1,062 unique images were repartitioned into 744 training, 161 validation, and 157 locked test images. Four CNN-based models were compared, with ResNet50 and EfficientNet-B0 selected for reliability evaluation. In the 157-image locked test set, ResNet50 correctly classified 151 images (96.18%) with a macro-F1 of 95.90%. Under strong image corruption, ResNet50 retained 91.52% macro-F1, compared to 83.40% for EfficientNet-B0. Temperature scaling reduced ResNet50's expected calibration error from 0.103 to 0.030. At approximately 90% selective coverage, it achieved 99.30% selective accuracy while rejecting 5 of its 6 original test errors (83.33%). Grad-CAM, Score-CAM and integrated failure analysis further identified shared and high-confidence residual failure cases.

2. Dataset Justification

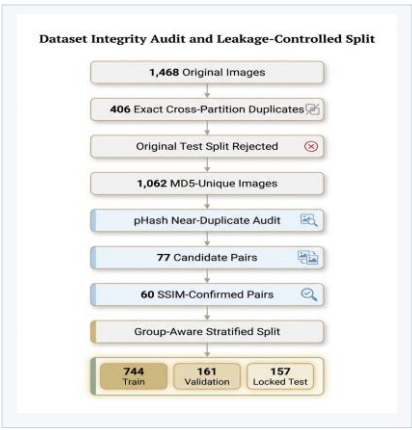
- Source & size:** Mendeley Data's *UAV-Based Aquatic Plant Dataset for Segmentation and Classification*, containing 1,468 images across 14 aquatic plant classes. After duplicate auditing, 1,062 unique images were retained.

Fig. 1. Aquatic Plant Dataset samples per class.



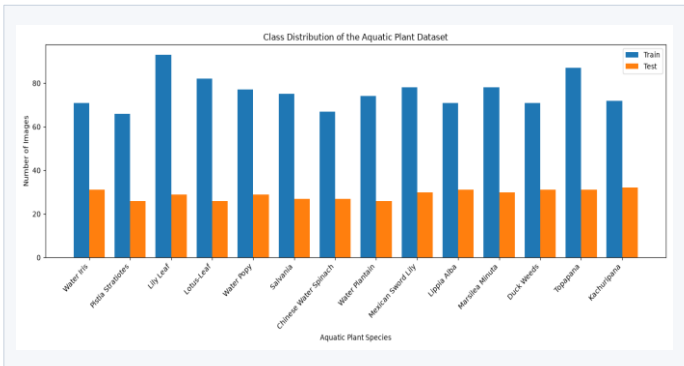
- Class distribution:** The classes were mildly imbalanced, with approximately 46–65 images per class (ratio $\approx 1.41:1$). No class-weighting was applied; standard cross-entropy loss was used because the imbalance was limited.
- Split:** The 1,062 unique images were divided using a group-aware stratified split into 744 training, 161 validation, and 157 locked-test images. Exact duplicate leakage was removed using MD5 hashing, while confirmed near-duplicates identified using pHash and SSIM were kept within the same partition.

Fig. 2. Dataset integrity audit & leakage-controlled split pipeline.



- Preprocessing:** Images were resized/cropped to 224×224 pixels and normalized using ImageNet mean and standard deviation. Training used random rotation, random resized cropping, horizontal flipping, and mild color jitter. Validation and test images used deterministic resizing and normalization.
- Fit:** The dataset contains natural UAV/acquisition images of aquatic vegetation, making ImageNet-pretrained CNN backbones suitable for transfer learning. ResNet50 and EfficientNet-B0 were therefore used to transfer general visual features to the aquatic plant classification task.

Fig. 3. 14 Aquatic plant classes distribution.



3. Methodology: Transfer Learning

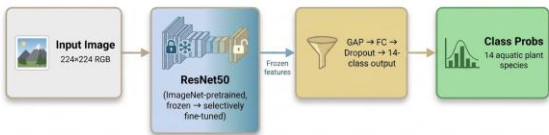
Fig. 4. framework of the proposed reliability-aware aquatic plant classification pipeline.



Four backbones compared: custom CNN, VGG16, ResNet50, EfficientNet-B0 (ImageNet-pretrained). ResNet50 selected for highest validation macro-F1 (98.13%), best locked-test accuracy (96.18%), strongest corruption robustness (91.52% macro-F1), best calibration (ECE 0.030), and best selective prediction (99.30% accuracy, 83.33% error rejection).

Fig. 5. Architecture: frozen backbone + modified head.

Architecture Diagram



MODEL SELECTION & MODIFICATION

ResNet50 selected via validation macro-F1 (98.13%), outperforming VGG16 (97.50%), EfficientNet-B0 (97.52%), and custom CNN (96.89%). Backbone frozen during initial comparison; upper layers selectively unfrozen during fine-tuning, using a smaller LR for pretrained weights and higher LR for the new head. Original 1000-class ImageNet head replaced with a 14-class GAP→FC→Dropout head.

JUSTIFICATION

ResNet50 had the highest validation macro-F1 and later proved most robust under corruption (91.52% vs. 83.40% macro-F1 at strongest severity) and best calibrated (ECE 0.030 vs. 0.043 post-scaling) confirming it as the stronger choice beyond clean accuracy alone.

4. Augmentation, Training & Fine-tuning

Augmentation: Random rotation ($\pm 10^\circ$), random resized cropping (scale 0.85–1.0), horizontal flipping, and mild brightness/contrast/saturation jitter applied to training images only, to increase robustness to UAV viewpoint, scale, and lighting variation without distorting plant morphology. Validation and test images used only deterministic resizing (224×224) and ImageNet normalization to ensure consistent, unbiased evaluation.

Training: Optimizer: AdamW. Loss: cross-entropy. Learning rate: reduced on plateau, with a smaller LR applied to pretrained backbone parameters and a higher LR for the new classifier head during fine-tuning. Epochs: up to 100, with early stopping (patience = 10 epochs) based on validation macro-F1. Batch size: 32. Random seed: fixed at 42 throughout

Fig. 6. Augmentation pipeline.

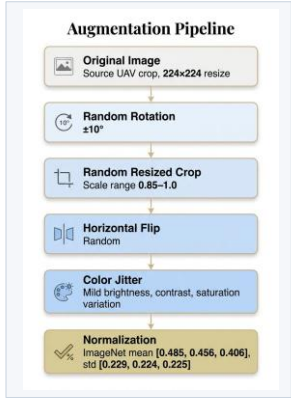
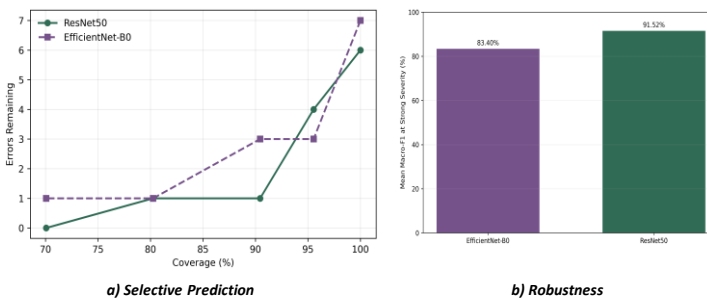


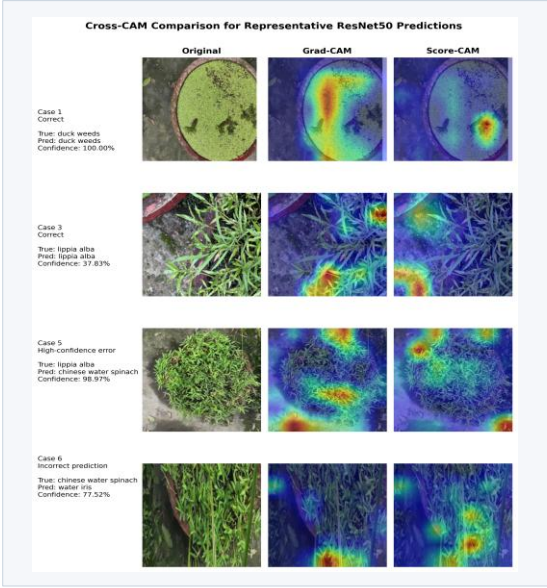
Fig. 7. Reliability comparison of the final models



5. Grad-CAM Visualizations & Interpretation

Grad-CAM was applied to ResNet50's final convolutional block (layer4[-1]) and EfficientNet-B0's final feature block (features[-1]) to visualize class-discriminative regions driving each prediction. Score-CAM is a gradient-free, activation-based alternative was additionally applied to ResNet50 for cross-validation of attention patterns, since it was computationally feasible only for the best-performing model. Representative cases were selected to span high-confidence correct predictions, low-confidence correct predictions, and high-confidence errors, allowing model attention to be examined across different reliability conditions rather than only on successful cases.

Fig. 8. Cross-CAM comparison of Grad-CAM and Score-CAM explanations.



Interpretation Note Grad-CAM and Score-CAM attention concentrates on plant canopy and leaf-texture regions in correct predictions, but grows diffuse and spreads across competing background vegetation in uncertain or erroneous cases — most notably in the recurring Lippia alba → Chinese water spinach confusion, where high confidence ($\sim 98.9\%$) coincided with ambiguous, non-discriminative attention

6. Evaluation Metrics & Comparison

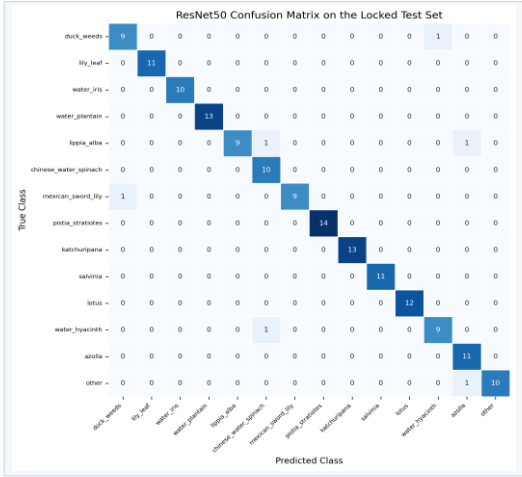
Table 1. VALIDATION-BASED MODEL SELECTION AND LOCKED-TEST PERFORMANCE.

Model	Val. F1 (%)	Selected	Test Acc. (%)	Test F1 (%)	AUC
Custom CNN	96.89	No	93.24	92.86	0.9901
VGG16	97.50	No	94.67	94.12	0.9953
EfficientNet-B0	97.52	Yes	95.54	95.21	0.9974
ResNet50	98.13	Yes	96.18	95.90	0.9995

Table 2. RELIABILITY RESULTS OF THE FINAL MODELS.

Model	Mean Corr. F1 (%)	Strong F1 (%)	ECE B/A	Sel. Acc. (%)	Err. Rej. (%)
EfficientNet-B0	90.62	83.40	0.057/0.043	97.89	57.14
ResNet50	92.80	91.52	0.103/0.030	99.30	83.33

Fig. 9 Confusion matrix of the fine-tuned ResNet50 on the 157-image locked test set .



The confusion matrix shows ResNet50's errors concentrated among visually similar species, most notably Lippia alba misclassified as Chinese water spinach — a confusion that occurred across both architectures with high calibrated confidence ($\sim 98.9\%$), suggesting genuine visual ambiguity rather than a model-specific weakness. Additional minor confusions include Mexican sword lily and duck weeds.

7. References & Bonus

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[2] M. Tan, Q. V. Le. "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks." ICML, 2019.
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